SpectSPD-DANN: Riemannian Geometry and Adversarial SPD Alignment for Cross-Dataset EEG Depression Recognition

Author One, *Member, IEEE,* Author Two, and Author Three

TABLE I
EEG DEPRESSION DATASETS

Dataset	N	MDD/HC	Ch.	f_s (Hz)	Label
MODMA [1]	53	24/29	128	1000	Clinical
Mumtaz [2]	58	34/24	14	256	Clinical
OpenNeuro [3]	121	42/79	64	256	BDI ≥ 14
Total	**232**	**100/132**	—	—	—

***Abstract*—ABSTRACT PLACEHOLDER — DO NOT SUBMIT.**

***Index Terms*—electroencephalography, depression detection, Riemannian geometry, symmetric positive definite matrices, domain adaptation, transfer learning**

I. INTRODUCTION

II. RELATED WORK

III. METHODOLOGY

The proposed SpectSPD-DANN is a Riemannian geometry framework for EEG-based depression detection that operates on the manifold of symmetric positive-definite (SPD) matrices throughout the pipeline, from covariance estimation to cross-dataset domain alignment. Let $\mathcal{D}_k = \{(\mathbf{X}_{k,i},\, y_{k,i})\}_{i=1}^{N_k}$ denote the k-th EEG dataset, where $\mathbf{X}_{k,i} \in \mathbb{R}^{p\times T}$ is a multi-channel segment with p electrodes and T samples (columns corresponding to time points), and $y_{k,i} \in \{0,1\}$ indicates whether the subject belongs to the MDD or HC group. The corresponding SPD covariance representation is derived from $\mathbf{X}_{k,i}$ in the processing stages described below. Depending on the available training data, SpectSPD-DANN operates in two modes selected according to an empirically observed performance crossover as dataset size increases: for single-dataset evaluation at clinical scale ($N \approx 53$), a transductive Riemannian nearest-neighbour mode (NrDE) is used; for cross-dataset transfer and pooled evaluation, the full deep adversarial SPD encoder is engaged.

A. Datasets

This study uses three publicly available resting-state EEG depression datasets. All original studies were conducted under institutional review board oversight with written informed consent; IRB details are reported in the respective publications [1]–[3]. Table I summarises the study characteristics.

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Corresponding author: TODO (email: TODO@institution.edu).

OpenNeuro assigns labels based on BDI self-report scores (BDI ≥ 14 indicates MDD), whereas MODMA and Mumtaz use clinician-administered structured interviews, introducing a labeling-protocol mismatch whose impact is analysed separately in Section IV. Cross-dataset evaluation uses leave-one-dataset-out (LODO): train on $K-1=2$ datasets, test on all subjects of the held-out K-th; with $K=3$, this yields three LODO folds. Within-dataset evaluation uses 10-fold subject-stratified cross-validation within each $\mathcal{D}_k$; test subjects appear in no training fold.

B. EEG Covariance Representation

The pairwise statistical relationships between EEG channels, which partially reflect functional interactions between cortical regions, are a recognised biomarker for major depression: inter-hemispheric asymmetry and fronto-parietal coherence changes have been reported consistently across studies [4], [5]. A covariance matrix $\mathbf{C} \in \mathbb{R}^{p\times p}$ captures second-order pairwise linear relationships between channels; treating it as a geometric object on the SPD manifold, rather than vectorising it into a $\frac{p(p+1)}{2}$-dimensional Euclidean representation, preserves congruence-transform invariance that makes the features robust to between-session amplitude scaling and electrode impedance variation.

Let $\mathbf{X} \in \mathbb{R}^{p\times T}$ denote a 30-second EEG epoch with $p = 17$ harmonised channels and columns corresponding to time samples. The mean-centred sample covariance is

$$\hat{\mathbf{C}} = \frac{1}{T-1}\,\mathbf{X}\mathbf{H}\mathbf{X}^{\top}, \tag{1}$$

where $\mathbf{H} = \mathbf{I}_T - \frac{1}{T}\mathbf{1}_T\mathbf{1}_T^{\top}$ is the $T\times T$ centering matrix. Although $T \gg p$ in this setting, shrinkage is applied to improve numerical conditioning and robustness. We use the Ledoit-Wolf analytic shrinkage estimator [6], which minimises the expected Frobenius loss to the true population covariance:

$$\mathbf{C} = (1-\rho)\,\hat{\mathbf{C}} + \rho\,\hat{\mu}\,\mathbf{I}_p, \quad \hat{\mu} = \frac{\mathrm{tr}(\hat{\mathbf{C}})}{p}, \tag{2}$$

where $\rho \in [0,1]$ is the analytically optimal shrinkage coefficient and $\hat{\mu}$ is the average channel variance used as the shrinkage target. For the estimated $\rho > 0$, this yields a strictly positive-definite matrix $\mathbf{C} \succ \mathbf{0}$, a necessary precondition for Riemannian operations.

Depression alters EEG power and coherence differently across frequency bands. The raw epoch is bandpass-filtered into $B = 5$ commonly used EEG frequency bands: δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz), and γ (30–45 Hz), following the protocol of Barachant et al. [7]. A separate regularised covariance $\mathbf{C}^{(b)} \in \mathcal{S}_p^{++}$ is estimated in

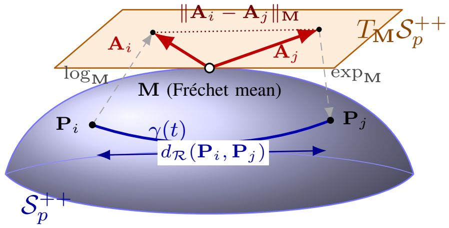

Fig. 1. Geometry of the SPD manifold $\mathcal{S}_p^{++}$. Two covariance matrices $\mathbf{P}_i, \mathbf{P}_j$ lie on the curved manifold and are joined by the geodesic $\gamma(t)$, whose length is the affine-invariant Riemannian distance $d_{\mathcal{R}}(\mathbf{P}_i, \mathbf{P}_j)$ (5). The tangent space $T_{\mathbf{M}}\mathcal{S}_p^{++}$ is attached to the manifold at the Fréchet mean $\mathbf{M}$, which serves as the origin of the tangent space. The logarithmic map (8) maps each point to a tangent vector anchored at $\mathbf{M}$, $\mathbf{A}_i = \mathrm{Log}_{\mathbf{M}}(\mathbf{P}_i)$ and $\mathbf{A}_j = \mathrm{Log}_{\mathbf{M}}(\mathbf{P}_j)$; the exponential map (7) is its inverse in a neighbourhood of $\mathbf{M}$. Once in the flat tangent space, points are compared by the Euclidean norm $\|\mathbf{A}_i - \mathbf{A}_j\|_{\mathbf{M}}$, which agrees with $d_{\mathcal{R}}$ to first order near $\mathbf{M}$ and enables standard vector operations on the resulting features.

each band $b \in \{1, \ldots, B\}$ via (2). This band-specific representation captures frequency-resolved connectivity structure that a broadband covariance would conflate.

Prior to covariance estimation, each recording is bandpass-filtered (0.5–45 Hz, 4th-order Butterworth, zero-phase), common-average referenced, downsampled to 256 Hz, and segmented into non-overlapping 30-second epochs. Epochs with peak-to-peak artefacts exceeding 150 μV are discarded. The 17-channel harmonised montage (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz), the maximal common 10-20 subset across all three datasets, is applied before covariance estimation.

C. Geometry of the SPD Manifold

Covariance matrices are not general real matrices: they must be symmetric and positive definite. Although $\mathcal{S}_p^{++}$ is an open subset of Sym_p, the vector space of $p \times p$ symmetric matrices, its natural geometry is non-Euclidean [8], [9]. Performing Euclidean operations on this space, such as computing the arithmetic mean or measuring proximity by Frobenius norm, ignores the intrinsic manifold geometry and may lead to distorted statistical estimates; Euclidean interpolation does not follow geodesics and can distort distances and averages. Riemannian geometry provides intrinsic distance, tangent vectors, and geodesics that respect the positive-definite structure, as illustrated in Fig. 1.

The SPD manifold of order p is defined as

$$\mathcal{S}_p^{++} = \left\{ \mathbf{C} \in \mathbb{R}^{p \times p} : \mathbf{C} = \mathbf{C}^\top,\ \mathbf{v}^\top \mathbf{C} \mathbf{v} > 0 \ \forall \mathbf{v} \in \mathbb{R}^p \setminus \{\mathbf{0}\} \right\}. \tag{3}$$

$\mathcal{S}_p^{++}$ is open in Sym_p with respect to the topology induced on Sym_p (it is not open in $\mathbb{R}^{p \times p}$), and is therefore a smooth manifold of dimension $\frac{p(p+1)}{2}$. At each point $\mathbf{P} \in \mathcal{S}_p^{++}$, the tangent space is identified with $T_{\mathbf{P}}\mathcal{S}_p^{++} = \mathrm{Sym}_p$ [10].

The manifold is endowed with the affine-invariant Riemannian metric (AIRM) [8], whose inner product at $\mathbf{P}$ is

$$\langle \mathbf{U}, \mathbf{V} \rangle_{\mathbf{P}} = \mathrm{tr}\big(\mathbf{P}^{-1} \mathbf{U}\, \mathbf{P}^{-1} \mathbf{V}\big), \quad \mathbf{U}, \mathbf{V} \in T_{\mathbf{P}}\mathcal{S}_p^{++}. \tag{4}$$

The induced norm on the tangent space is $\|\mathbf{S}\|_{\mathbf{P}} = \sqrt{\langle \mathbf{S}, \mathbf{S} \rangle_{\mathbf{P}}} = \sqrt{\mathrm{tr}(\mathbf{P}^{-1}\mathbf{S}\mathbf{P}^{-1}\mathbf{S})}$ for $\mathbf{S} \in T_{\mathbf{P}}\mathcal{S}_p^{++}$. The geodesic distance induced by the AIRM between $\mathbf{A}, \mathbf{B} \in \mathcal{S}_p^{++}$ is

$$d_{\mathcal{R}}(\mathbf{A}, \mathbf{B}) = \big\| \mathrm{logm}\big(\mathbf{A}^{-\frac{1}{2}} \mathbf{B} \mathbf{A}^{-\frac{1}{2}}\big) \big\|_F = \sqrt{\sum_{i=1}^{p} \log^2 \lambda_i(\mathbf{A}^{-1}\mathbf{B})}, \tag{5}$$

where $\mathrm{logm}(\cdot)$ is the principal matrix logarithm, $\|\cdot\|_F$ is the Frobenius norm, and $\lambda_i(\mathbf{A}^{-1}\mathbf{B})$ denotes the i-th eigenvalue of $\mathbf{A}^{-1}\mathbf{B}$, equivalently the generalised eigenvalues of the pair $(\mathbf{B}, \mathbf{A})$ solving $\mathbf{B}\mathbf{v} = \lambda \mathbf{A}\mathbf{v}$. Although $\mathbf{A}^{-1}\mathbf{B}$ is generally non-symmetric, it is similar to the SPD matrix $\mathbf{A}^{-1/2}\mathbf{B}\mathbf{A}^{-1/2}$, so its eigenvalues are real and strictly positive and the logarithm in (5) is well defined. The AIRM is invariant under congruence transformations $(\mathbf{A}, \mathbf{B}) \mapsto (\mathbf{W}\mathbf{A}\mathbf{W}^\top, \mathbf{W}\mathbf{B}\mathbf{W}^\top)$ for any invertible $\mathbf{W}$ [8], making the metric insensitive to invertible linear reparameterizations of the signal space. In the EEG setting, per-channel gain and session-level amplitude scaling correspond to diagonal choices of $\mathbf{W}$ and are thus special cases of this invariance [7].

For $\mathbf{P}, \mathbf{Q} \in \mathcal{S}_p^{++}$, the unique AIRM geodesic joining them is

$$\gamma(t) = \mathbf{P}^{\frac{1}{2}} \big(\mathbf{P}^{-\frac{1}{2}} \mathbf{Q} \mathbf{P}^{-\frac{1}{2}}\big)^t \mathbf{P}^{\frac{1}{2}}, \quad t \in [0, 1], \tag{6}$$

with $\gamma(0) = \mathbf{P}$ and $\gamma(1) = \mathbf{Q}$ [9]. The exponential map $\mathrm{Exp}_{\mathbf{P}} : T_{\mathbf{P}}\mathcal{S}_p^{++} \to \mathcal{S}_p^{++}$ maps a tangent vector $\mathbf{S}$ to a manifold point by following this geodesic from $\mathbf{P}$ in direction $\mathbf{S}$:

$$\mathrm{Exp}_{\mathbf{P}}(\mathbf{S}) = \mathbf{P}^{\frac{1}{2}} \mathrm{expm}\big(\mathbf{P}^{-\frac{1}{2}} \mathbf{S} \mathbf{P}^{-\frac{1}{2}}\big) \mathbf{P}^{\frac{1}{2}}. \tag{7}$$

Its inverse, the logarithmic map $\mathrm{Log}_{\mathbf{P}} : \mathcal{S}_p^{++} \to T_{\mathbf{P}}\mathcal{S}_p^{++}$, maps a second SPD matrix $\mathbf{Q} \in \mathcal{S}_p^{++}$ back to the tangent space and is

$$\mathrm{Log}_{\mathbf{P}}(\mathbf{Q}) = \mathbf{P}^{\frac{1}{2}} \mathrm{logm}\big(\mathbf{P}^{-\frac{1}{2}} \mathbf{Q} \mathbf{P}^{-\frac{1}{2}}\big) \mathbf{P}^{\frac{1}{2}}, \tag{8}$$

and satisfies $d_{\mathcal{R}}(\mathbf{P}, \mathbf{Q}) = \|\mathrm{Log}_{\mathbf{P}}(\mathbf{Q})\|_{\mathbf{P}}$ [10]. The pair $(\mathrm{Exp}_{\mathbf{P}}, \mathrm{Log}_{\mathbf{P}})$ enables transfer of manifold computations to a locally isometric Euclidean space.

The Frechet mean (Riemannian centre of mass) of a set $\{\mathbf{C}_i\}_{i=1}^N \subset \mathcal{S}_p^{++}$ is

$$\mathbf{M}^* = \arg\min_{\mathbf{M} \in \mathcal{S}_p^{++}} \sum_{i=1}^{N} d_{\mathcal{R}}^2(\mathbf{M}, \mathbf{C}_i), \tag{9}$$

which exists and is unique: $(\mathcal{S}_p^{++}, d_{\mathcal{R}})$ is a Hadamard manifold (complete, simply connected, of non-positive sectional curvature), so the squared-distance objective in (9) is geodesically convex and admits a single minimiser [9], [11]. It is computed by the iterative Karcher mean update, a Riemannian gradient descent on (9),

$$\mathbf{M}_{k+1} = \mathrm{Exp}_{\mathbf{M}_k}\Big(\frac{1}{N} \sum_{i=1}^{N} \mathrm{Log}_{\mathbf{M}_k}(\mathbf{C}_i)\Big), \tag{10}$$

which converges to the unique Fréchet mean; a few iterations suffice in practice [12].

D. Riemannian Feature Extraction

The logarithmic map at the Frechet mean $\mathbf{M}^*$ provides a locally isometric embedding of $\mathcal{S}_p^{++}$ into the flat tangent space $T_{\mathbf{M}^*}\mathcal{S}_p^{++} \cong \mathrm{Sym}_p$. Choosing $\mathbf{M}^*$ as the expansion point minimises the average linearisation error across the training set, yielding the most faithful Euclidean approximation to the local Riemannian geometry of the data [13], [14].

Given the per-band Frechet mean $\mathbf{M}^{(b)*}$ computed from training-fold covariance matrices $\{\mathbf{C}_i^{(b)}\}_{i\in\mathcal{I}_{\text{train}}}$ via (10), each covariance $\mathbf{C}^{(b)}$ is mapped to a symmetric tangent matrix:

$$\mathbf{S}^{(b)} = \mathrm{Log}_{\mathbf{M}^{(b)*}}\big(\mathbf{C}^{(b)}\big) = \mathbf{M}^{(b)*\frac{1}{2}}\mathrm{logm}\big(\mathbf{M}^{(b)*^{-\frac{1}{2}}}\mathbf{C}^{(b)}\,\mathbf{M}^{(b)*^{-\frac{1}{2}}}\big)\mathbf{M}^{(b)*\frac{1}{2}} \tag{11}$$

The upper triangle (including diagonal) of $\mathbf{S}^{(b)}$ is vectorised to

$$\mathbf{s}^{(b)} = \mathrm{vech}\big(\mathbf{S}^{(b)}\big) \in \mathbb{R}^{p(p+1)/2}, \tag{12}$$

where $\mathrm{vech}(\cdot)$ stacks the upper-triangular entries column by column. With $p = 17$, each band yields a 153-dimensional descriptor. The mean $\mathbf{M}^{(b)*}$ is computed on training data only and applied to held-out test data without update, preventing data leakage.

To complement the spatial connectivity structure in $\mathbf{s}^{(b)}$, spectral power information is incorporated via differential entropy [15]. For band b, the differential entropy of a Gaussian with variance $\hat{\sigma}_b^2$ is

$$\mathrm{DE}^{(b)} = \frac{1}{2}\log\big(2\pi e\,\hat{\sigma}_b^2\big), \tag{13}$$

where $\hat{\sigma}_b^2$ is the per-channel averaged band-filtered signal variance. The combined NrDE feature vector is

$$\mathbf{f} = \big[\mathbf{s}^{(1)},\ldots,\mathbf{s}^{(B)},\mathrm{DE}^{(1)},\ldots,\mathrm{DE}^{(B)}\big] \in \mathbb{R}^{B\cdot\frac{p(p+1)}{2}+B}, \tag{14}$$

yielding a 770-dimensional representation per epoch ($B = 5$, $p = 17$: $5\times153+5=770$).

For within-dataset evaluation, SpectSPD-DANN operates in its transductive Riemannian nearest-neighbour mode. Rather than training a global classifier on all subjects, for each test window with per-band covariances $\{\mathbf{C}_{\text{test}}^{(b)}\}_{b=1}^{B}$, all training subjects $j \in \mathcal{I}_{\text{train}}$ are ranked by their mean Riemannian distance to the test point:

$$r_j = \frac{1}{B}\sum_{b=1}^{B} d_{\mathcal{R}}\big(\mathbf{C}_{\text{test}}^{(b)},\,\mathbf{M}_j^{(b)}\big), \tag{15}$$

where $\mathbf{M}_j^{(b)}$ is the Frechet mean of subject j's per-band covariance matrices. The $K = 3$ subjects with the smallest r_j are selected, and a logistic regression classifier (ℓ_2-regularised, $C_{\text{reg}} = 1$) is fit on their windows [16]. This ensures the classifier is conditioned on subjects whose neural synchrony geometry is maximally similar to the test subject. $K = 3$ was selected by a one-time sweep over $K \in \{1,2,3,4,5,8\}$ on a held-out validation partition and frozen across all datasets and folds.

E. Domain Adaptation on SPD Manifolds

When training on source dataset $\mathcal{D}_s$ and testing on target $\mathcal{D}_t$, the EEG covariance distributions differ: $P_s(\mathbf{C}) \neq P_t(\mathbf{C})$ even when the class-conditional distribution $P(y \mid \mathbf{C})$ is approximately invariant. This covariate shift on $\mathcal{S}_p^{++}$ arises from hardware differences, impedance variation, and inter-site demographic factors. The shift manifests as a rotation of the principal axes of the covariance distribution, not a simple location offset: re-centring at the Riemannian mean corrects the first-order shift but leaves residual directional misalignment that degrades classification. Adversarial domain alignment is therefore preferred over re-centring approaches [17].

Let the source domain contain N_s labelled windows $\{(\mathbf{X}_i^s, y_i)\}_{i=1}^{N_s}$ and the target domain contain N_t unlabelled windows $\{\mathbf{X}_j^t\}_{j=1}^{N_t}$ from the held-out dataset (transductive setting: target windows are present at training time, but never their labels). An encoder $F_e : \mathbb{R}^{p\times T} \to \mathbb{R}^d$, task classifier $F_c : \mathbb{R}^d \to [0,1]$, and domain discriminator $F_d : \mathbb{R}^d \to [0,1]$ are trained jointly as a min-max game:

$$\min_{\theta_e,\theta_c}\max_{\theta_d}\ \mathcal{L}_{\text{cls}}(\theta_e,\theta_c)\ -\ \lambda\,\mathcal{L}_{\text{dom}}(\theta_e,\theta_d), \tag{16}$$

where θ_e, θ_c, θ_d are the respective parameter sets and $\lambda > 0$ controls the trade-off between task accuracy and domain alignment.

F. Proposed Network Architecture

SpectSPD-DANN consists of two parallel feature streams, an SPD manifold branch and a spectral DE branch, whose representations are fused before the task classifier and the domain discriminator. Figure 2 illustrates the complete architecture.

The SPD branch maps a raw EEG segment $\mathbf{X} \in \mathbb{R}^{p\times T}$ to a manifold-aware Euclidean embedding through four sequential stages.

Stage 1: Temporal convolution. A 1-D causal convolution with n_f filters of length ℓ maps $\mathbf{X}$ to a hidden representation $\mathbf{H} \in \mathbb{R}^{p\times T'}$. This step learns band-selective spatial filters jointly with the downstream SPD objective, adapting to the depression-relevant frequency structure in the data.

Stage 2: Differentiable covariance pooling. A symmetric covariance pooling layer collapses the temporal axis:

$$\mathbf{C}_0 = \frac{1}{T'}\mathbf{H}\mathbf{H}^\top \ \in\ \mathcal{S}_p^{+}. \tag{17}$$

This differentiable operation, used in SPDNet-style architectures [18], [19], makes the branch end-to-end trainable while operating in $\mathcal{S}_p^{++}$ from this stage forward.

Stage 3: BiMap projection. The $p \times p$ SPD matrix $\mathbf{C}_0$ is projected to a lower-dimensional $q \times q$ SPD matrix via a learnable congruence transformation:

$$\mathbf{C}_1 = \phi_{\mathbf{W}}(\mathbf{C}_0) = \mathbf{W}\mathbf{C}_0\mathbf{W}^\top, \quad \mathbf{W} \in \mathbb{R}^{q\times p}, \quad q = 12. \tag{18}$$

If $\mathbf{W}$ has full row rank, then for any $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v}^\top\mathbf{C}_1\mathbf{v} = (\mathbf{W}^\top\mathbf{v})^\top\mathbf{C}_0(\mathbf{W}^\top\mathbf{v}) > 0$, so $\mathbf{C}_1 \in \mathcal{S}_q^{++}$. This BiMap layer, introduced in SPDNet [18], is the canonical manifold-preserving dimensionality reduction on $\mathcal{S}^{++}$; it reduces the $p=17$ covariance to a $q=12$ subspace that concentrates the depression-relevant variance.

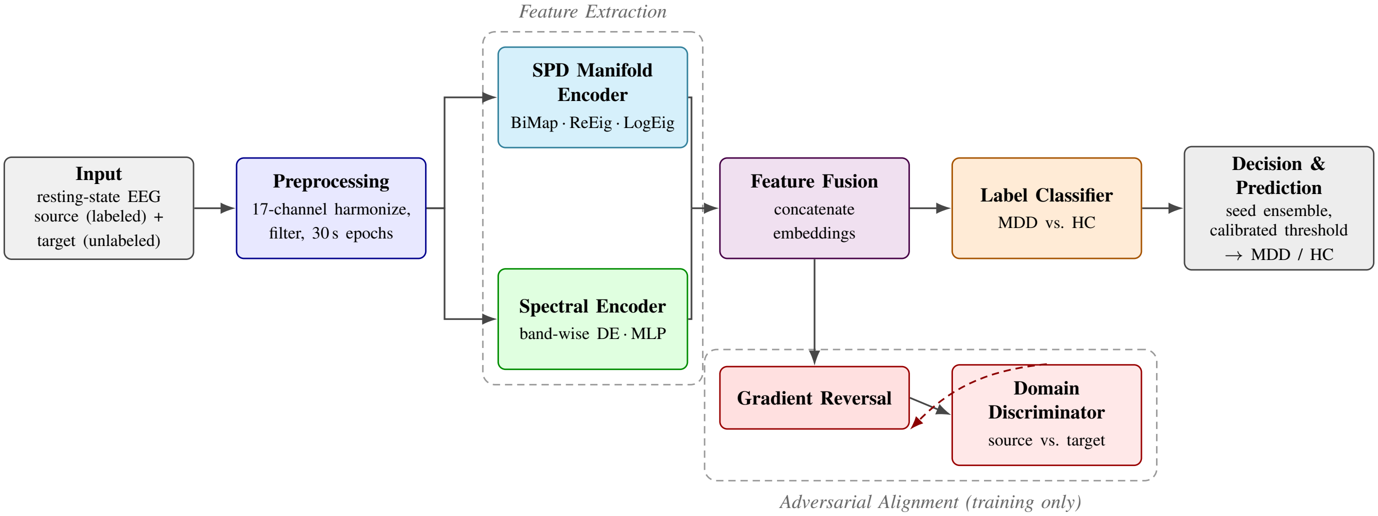

Fig. 2. Architecture of the proposed SpectSPD-DANN framework. Harmonized resting-state EEG from the labeled source corpora and the unlabeled held-out target corpus is encoded by two parallel branches: an SPD manifold encoder (covariance pooling followed by BiMap, ReEig, and LogEig layers) and a spectral encoder (band-wise differential entropy with an MLP). The fused representation feeds a label classifier trained on source labels and, through a gradient reversal layer, a domain discriminator that adversarially aligns the source and target distributions during training. At inference the discriminator branch is discarded; subject-level scores are combined by a seed ensemble and thresholded with a calibrated decision rule.

Stage 4: ReEig and LogEig. Gradient-based optimisation can drive eigenvalues of $\mathbf{C}_1$ toward zero, violating strict positive definiteness. The rectified eigenvalue (ReEig) layer [18] floors all eigenvalues at ε:

$$\mathbf{C}_2 = \phi_{\text{rec}}(\mathbf{C}_1) = \mathbf{U}\,\text{diag}\big(\max(\lambda_1,\varepsilon),\ldots,\max(\lambda_q,\varepsilon)\big)\mathbf{U}^\top, \tag{19}$$

where $\mathbf{C}_1 = \mathbf{U}\,\text{diag}(\lambda_1,\ldots,\lambda_q)\,\mathbf{U}^\top$ is the eigendecomposition and $\varepsilon = 10^{-4}$. The matrix logarithm (LogEig) layer then maps $\mathbf{C}_2$ to the tangent space at the identity $\mathbf{I}_q$, following the log-Euclidean framework [20]:

$$\mathbf{Z} = \phi_{\log}(\mathbf{C}_2) = \text{logm}(\mathbf{C}_2) = \mathbf{U}\,\text{diag}(\log\lambda_1,\ldots,\log\lambda_q)\,\mathbf{U}^\top, \tag{20}$$

vectorised to $\mathbf{z}_{\text{spd}} = \text{vech}(\mathbf{Z}) \in \mathbb{R}^{q(q+1)/2}$. With $q = 12$, the SPD branch produces a 78-dimensional output. ReEig and LogEig have no learnable parameters; their Jacobians are computed analytically for backpropagation through the eigendecomposition [18].

The DE spectral features from (13) are computed over the same epoch in the same $B = 5$ bands and passed through a two-layer MLP with ReLU activations, producing $\mathbf{z}_{\text{de}} \in \mathbb{R}^{d_{\text{de}}}$. The two branch outputs are then concatenated to form the shared representation:

$$\mathbf{h} = \big[\mathbf{z}_{\text{spd}};\,\mathbf{z}_{\text{de}}\big] \in \mathbb{R}^d, \quad d = \tfrac{q(q+1)}{2} + d_{\text{de}}, \tag{21}$$

which is passed to both the task head F_c and the domain discriminator F_d.

G. Training Objective and Optimization

The task head F_c maps $\mathbf{h}$ to a probability $\hat{p} = F_c(\mathbf{h}) \in [0,1]$ via a fully connected layer with sigmoid activation. The binary cross-entropy over labelled source windows is

$$\mathcal{L}_{\text{cls}}(\theta_e,\theta_c) = -\frac{1}{N_s}\sum_{i=1}^{N_s}\big[y_i\log\hat{p}_i + (1-y_i)\log(1-\hat{p}_i)\big]. \tag{22}$$

The domain discriminator F_d maps $\mathbf{h}$ to a domain label probability $\hat{d} \in [0,1]$ ($d = 0$: source; $d = 1$: target), with cross-entropy

$$\mathcal{L}_{\text{dom}}(\theta_e,\theta_d) = -\frac{1}{N_s+N_t}\sum_{i=1}^{N_s+N_t}\big[d_i\log\hat{d}_i + (1-d_i)\log(1-\hat{d}_i)\big]. \tag{23}$$

The min-max problem (16) is solved with a single forward-backward pass using a gradient reversal layer (GRL) [17] between F_e and F_d. In the forward pass the GRL acts as the identity; in the backward pass it scales and negates the gradient flowing from F_d to F_e:

$$\frac{\partial\tilde{\mathcal{L}}}{\partial\mathbf{h}} = -\lambda\,\frac{\partial\mathcal{L}_{\text{dom}}}{\partial\mathbf{h}}. \tag{24}$$

The encoder is thus trained to simultaneously minimise $\mathcal{L}_{\text{cls}}$ and confuse the domain discriminator, yielding the joint objective:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} - \lambda\,\mathcal{L}_{\text{dom}}. \tag{25}$$

The coupling coefficient λ is annealed following the schedule of Ganin et al. [17]: $\lambda(t) = \frac{2}{1+\exp(-\gamma t/T_{\max})} - 1$, with $\gamma = 10$. All parameters are optimised jointly with Adam at learning rate 10^{-2} for 80 epochs, batch size 32. The BiMap weight $\mathbf{W}$ is initialised from a truncated normal distribution and regularised by its Frobenius norm to maintain full row rank.

Because the adversarial objective (25) is sensitive to initialisation of θ_d, four independent training runs are performed per LODO fold with seeds $\mathcal{S} = \{42, 1, 2, 3\}$. Per-window prediction scores from each run are averaged at the subject level:

$$\hat{p}_i^{\text{ens}} = \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}}\hat{p}_i^{(s)}, \tag{26}$$

where $\hat{p}_i^{(s)}$ is the subject-level mean score from seed s.

H. Inference Procedure

For within-dataset evaluation, each test subject's 30-second windows are evaluated independently via (15) and (14). The $K = 3$ nearest training subjects are located in Riemannian space, a logistic classifier is fit on their windows, and per-window scores are averaged to obtain a subject-level classification probability. For cross-dataset evaluation, the trained encoder F_e and task head F_c are applied to all windows of the held-out target dataset, with subject-level scores aggregated across seeds via (26).

A decision threshold τ^* is set by maximising balanced accuracy on the training-fold subjects:

$$\tau^* = \arg\max_{\tau \in \mathbb{R}} \mathrm{BAcc}\big(\{\hat{p}_i^{\mathrm{ens}}\}_{i \in \mathcal{I}_{\mathrm{train}}}, \tau\big), \tag{27}$$

where $\mathcal{I}_{\mathrm{train}}$ indexes training-fold subjects. This threshold is then applied unchanged to the held-out test subjects. Fitting τ^* on the test set itself inflates accuracy by an amount proportional to the test set size and is explicitly avoided.

The primary metric is AUC (area under the ROC curve) [21], which measures $P(\hat{p}_{\mathrm{MDD}} > \hat{p}_{\mathrm{HC}})$, is threshold-free, and is invariant to class imbalance. Balanced accuracy and accuracy at τ^* are secondary. Pairwise model comparisons use the Wilcoxon signed-rank test across $N = 10$ CV folds; all results follow the format: mean ± std (p-value; Wilcoxon signed-rank; $N = 10$ folds). Per-fold LODO values are reported individually.

IV. Experimental Results

V. Discussion

VI. Conclusion

Acknowledgments

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